

YIMO 2026

Finals Round

Window A Division 1

1. Let a, b, c be positive real numbers satisfying

$$a + b + c = \frac{1}{a} + \frac{1}{b} + \frac{1}{c} \quad \text{and} \quad a^2 + b^2 + c^2 = \frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2} = 6.$$

The value of $a + b + c$ can be expressed in the form $p + \sqrt{q}$ for positive integers p and q . Find $p + q$.

Answer: 8

Solution: Let $s_1 = a + b + c$, $s_2 = ab + bc + ca$, and $s_3 = abc$. The first equation states $s_1 = \frac{s_2}{s_3}$. From $a^2 + b^2 + c^2 = s_1^2 - 2s_2 = 6$ and

$$\frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2} = \frac{s_2^2 - 2s_1s_3}{s_3^2} = 6,$$

substitute $s_2 = s_1s_3$ into both. The two equations become $s_1^2 - 2s_1s_3 = 6$ and $s_1^2 - \frac{2s_1}{s_3} = 6$. Subtracting gives $2s_1s_3 = \frac{2s_1}{s_3}$, so $s_3^2 = 1$ and hence $s_3 = 1$ (as the numbers are positive). Then $s_2 = s_1$ and $s_1^2 - 2s_1 - 6 = 0$, giving $s_1 = 1 + \sqrt{7}$. Therefore $p + q = 1 + 7 = 8$.

2. In triangle ABC , $AB = 9$, $BC = 10$, and $CA = 11$. The incircle of ABC touches side BC at point X . Line AX meets the incircle again at a second point Y (distinct from X). The length XY^2 can be expressed as $\frac{m}{n}$ where m and n are relatively prime positive integers. Find $m + n$.

Answer: 2377

Solution: The semiperimeter is $s = 15$, giving tangent lengths $BX = s - b = 4$, $CX = s - c = 6$, and tangent length from A equal to $s - a = 5$. The power of A with respect to the incircle is $5^2 = 25$, so $AX \cdot AY = 25$.

By Stewart's theorem on cevian AX in triangle ABC ,

$$AB^2 \cdot CX + AC^2 \cdot BX - AX^2 \cdot BC = BC \cdot BX \cdot CX,$$

so $81 \cdot 6 + 121 \cdot 4 - 10AX^2 = 240$, which gives $AX^2 = 73$. Since Y lies between A and X with $AY = \frac{25}{AX}$,

$$XY = AX - AY = \frac{AX^2 - 25}{AX} = \frac{48}{\sqrt{73}},$$

so $XY^2 = \frac{2304}{73}$. As 73 is prime and does not divide $2304 = 2^8 \cdot 3^2$, the fraction is reduced. Therefore $m + n = 2304 + 73 = 2377$.

3. Determine, with proof, all pairs of positive integers (a, b) such that both $a^2 + b$ and $b^2 + a$ are perfect squares.

Answer: No such pairs exist.

Solution: Without loss of generality assume $a \leq b$. Then

$$b^2 < b^2 + a \leq b^2 + b < b^2 + 2b + 1 = (b + 1)^2,$$

so $b^2 + a$ lies strictly between the consecutive perfect squares b^2 and $(b + 1)^2$, and therefore cannot be a perfect square. The same argument applied to $a^2 + b$ in the case $a \geq b$ gives the symmetric contradiction. Hence no pair (a, b) satisfies both conditions.

4. The cells of an $n \times n$ array contain real numbers (not necessarily distinct), and the sum of all entries is positive. In one move, you may multiply every entry of a single row by -1 , or every entry of a single column by -1 . Prove that it is always possible, after finitely many moves, to reach an arrangement in which the sum of the entries in every row and every column is nonnegative.

Solution: There are only 2^{2n} possible arrangements reachable by these moves (each row and each column is either flipped or not). Among all reachable arrangements, choose one that maximizes the total sum S of all entries.

We claim this arrangement has all row sums and column sums nonnegative. Suppose some row had a negative sum $\sigma < 0$. Flipping that row negates every entry in it, changing the total sum by $-2\sigma > 0$, producing a reachable arrangement with strictly larger total sum. This contradicts the maximality of S . The same argument applies to any column with negative sum. Hence every row sum and column sum is nonnegative.

Finally, since the original total sum is positive and flips only ever increase S toward the maximum, the maximizing arrangement is reachable in finitely many moves.

YIMO 2026

Finals Round

Window A Division 2

1. Find the sum of all real solutions x to the equation

$$x^4 + (x - 2)^4 = 82.$$

Answer: 2

Solution: Substitute $u = x - 1$ to exploit the symmetry about $x = 1$. Then

$$(u + 1)^4 + (u - 1)^4 = 2(u^4 + 6u^2 + 1) = 82,$$

so $u^4 + 6u^2 - 40 = 0$. Treating this as a quadratic in u^2 , $(u^2 + 10)(u^2 - 4) = 0$, and since $u^2 \geq 0$ we take $u^2 = 4$. Thus $u = \pm 2$, giving $x = 3$ and $x = -1$. Their sum is 2.

2. Find the number of positive integers n with $1 \leq n \leq 1000$ such that $n^2 + n + 1$ is divisible by 7.

Answer: 286

Solution: Checking residues modulo 7, $n^2 + n + 1 \equiv 0 \pmod{7}$ exactly when $n \equiv 2$ or $n \equiv 4 \pmod{7}$. In the range $1 \leq n \leq 1000$, each of these residue classes contains 143 values ($n = 2, 9, \dots, 996$ and $n = 4, 11, \dots, 998$). The total is $143 + 143 = 286$.

3. In acute triangle ABC , $AB = 5$, $AC = 7$, and the altitude from A to side BC has length 4. The length BC can be expressed in the form $\sqrt{m} + \sqrt{n}$ where m and n are positive integers with $m \leq n$. Find $m + n$.

Answer: 42

Solution: Let H be the foot of the altitude from A . In right triangle ABH , $BH = \sqrt{AB^2 - AH^2} = \sqrt{25 - 16} = 3$. In right triangle ACH , $CH = \sqrt{AC^2 - AH^2} = \sqrt{49 - 16} = \sqrt{33}$. Because the triangle is acute, H lies on segment BC , so

$$BC = BH + CH = 3 + \sqrt{33} = \sqrt{9} + \sqrt{33}.$$

Thus $m = 9$, $n = 33$, and $m + n = 42$.

4. Eight people stand in a circle. Each person silently chooses a positive integer and writes it on a card. Then each person announces the sum of the two numbers written by their two immediate neighbors. The announced numbers, going around the circle in order, are exactly 1, 2, 3, 4, 5, 6, 7, 8. Determine, with proof, whether such a configuration is possible.

Answer: Not possible.

Solution: Label the chosen integers $a_1, a_2, \dots, a_8$ around the circle. The number announced by person i is the sum of their two neighbors, $a_{i-1} + a_{i+1}$ (indices taken modulo 8). Each a_j is a positive integer, so every announced value satisfies

$$a_{i-1} + a_{i+1} \geq 1 + 1 = 2.$$

But one of the announced numbers is 1, which is less than 2. This is impossible, so no such configuration exists.

YIMO 2026

Finals Round

Window B Division 1

1. Let a, b, c be positive real numbers. Prove that

$$\frac{a^2}{b+c} + \frac{b^2}{c+a} + \frac{c^2}{a+b} \geq \frac{a+b+c}{2},$$

and determine, with proof, all triples (a, b, c) for which equality holds.

Solution: By the Cauchy–Schwarz inequality in Engel (Titu) form,

$$\frac{a^2}{b+c} + \frac{b^2}{c+a} + \frac{c^2}{a+b} \geq \frac{(a+b+c)^2}{(b+c) + (c+a) + (a+b)} = \frac{(a+b+c)^2}{2(a+b+c)} = \frac{a+b+c}{2}.$$

Equality in the Engel form requires $\frac{a}{b+c} = \frac{b}{c+a} = \frac{c}{a+b}$. Setting the first two ratios equal and cross-multiplying gives $a(c+a) = b(b+c)$, i.e. $a^2 - b^2 + c(a-b) = 0$, so $(a-b)(a+b+c) = 0$. Since $a+b+c > 0$, this forces $a = b$; symmetrically $b = c$. Hence equality holds if and only if $a = b = c$.

2. In triangle ABC , $\angle BAC = 60^\circ$, $AB = 5$, and $AC = 8$. The angle bisector from A meets segment BC at D , and the perpendicular from D to line AB meets AB at E . The length DE^2 can be expressed as $\frac{m}{n}$ where m and n are relatively prime positive integers. Find $m+n$.

Answer: 1369

Solution: The angle bisector from A has length

$$AD = \frac{2 \cdot AB \cdot AC \cos(\angle A/2)}{AB + AC} = \frac{2 \cdot 5 \cdot 8 \cdot \cos 30^\circ}{13} = \frac{80 \cdot \frac{\sqrt{3}}{2}}{13} = \frac{40\sqrt{3}}{13}.$$

The bisector makes a 30° angle with AB , so in right triangle ADE (right angle at E),

$$DE = AD \sin 30^\circ = \frac{AD}{2} = \frac{20\sqrt{3}}{13}.$$

Therefore $DE^2 = \frac{1200}{169}$. Since $\gcd(1200, 169) = 1$, we have $m+n = 1200 + 169 = 1369$.

3. Let p be an odd prime. Prove that

$$\binom{2p}{p} \equiv 2 \pmod{p^2}.$$

Solution: Write

$$\binom{2p}{p} = \frac{(2p)!}{p!p!} = \frac{(p+1)(p+2)\cdots(2p)}{p!} = 2 \prod_{k=1}^{p-1} \frac{p+k}{k} = 2 \prod_{k=1}^{p-1} \left(1 + \frac{p}{k}\right),$$

where the factor 2 comes from the $k = p$ term, $\frac{2p}{p} = 2$. Expanding the product modulo p^2 and discarding all terms containing p^2 ,

$$\prod_{k=1}^{p-1} \left(1 + \frac{p}{k}\right) \equiv 1 + p \sum_{k=1}^{p-1} \frac{1}{k} \pmod{p^2}.$$

Pairing k with $p - k$,

$$\frac{1}{k} + \frac{1}{p-k} = \frac{p}{k(p-k)} \equiv 0 \pmod{p},$$

and since p is odd these $\frac{p-1}{2}$ pairs exhaust the sum, giving $\sum_{k=1}^{p-1} \frac{1}{k} \equiv 0 \pmod{p}$. Therefore the product is $\equiv 1 \pmod{p^2}$ after multiplying the residue $p \cdot 0$ back in, and

$$\binom{2p}{p} \equiv 2 \cdot 1 = 2 \pmod{p^2}.$$

4. A prototype Artemis lunar landing platform has five landing legs: four legs L_1, L_2, L_3, L_4 positioned at the vertices of a square (labeled in order around the square), and a fifth leg L_5 positioned at the center of the square. Each leg initially has length 10.

For how many ordered 5-tuples $(k_1, k_2, k_3, k_4, k_5)$ of nonnegative integers can engineers shorten leg L_i by k_i units for each $i = 1, 2, 3, 4, 5$ and still have the platform rest stably on a flat lunar surface?

(The platform is stable if all five footpads can simultaneously touch the ground. A final leg length of 0 is permitted.)

Answer: 451

Solution: The four corner footpads are coplanar if and only if the two diagonal pairs drop by equal totals, that is $k_1 + k_3 = k_2 + k_4$ (here L_1, L_3 and L_2, L_4 are the opposite corners). When the corners are coplanar, the center footpad sits at the average height of the four corners, so stability further requires

$$4k_5 = k_1 + k_2 + k_3 + k_4,$$

which forces $k_1 + k_2 + k_3 + k_4$ to be divisible by 4. Each admissible (k_1, k_2, k_3, k_4) then determines a unique valid k_5 , so it suffices to count 4-tuples in $\{0, 1, \dots, 10\}^4$ with $k_1 + k_3 = k_2 + k_4$ and this common value even.

For a fixed common value s , the number of pairs (k_1, k_3) with $k_1 + k_3 = s$ and $0 \leq k_1, k_3 \leq 10$ is $11 - |s - 10|$, and likewise for (k_2, k_4) . The condition $k_1 + k_2 + k_3 + k_4 = 2s \equiv 0 \pmod{4}$ means s is even. Summing $(11 - |s - 10|)^2$ over even $s \in \{0, 2, 4, \dots, 20\}$,

$$1^2 + 3^2 + 5^2 + 7^2 + 9^2 + 11^2 + 9^2 + 7^2 + 5^2 + 3^2 + 1^2 = 451.$$

YIMO 2026

Finals Round

Window B Division 2

1. The polynomial $x^4 - 4x^3 + 6x^2 - 4x - 2024$ has four roots r_1, r_2, r_3, r_4 (counted with multiplicity, possibly complex). Find the value of $r_1^4 + r_2^4 + r_3^4 + r_4^4$.

Answer: 8104

Solution: Notice that $x^4 - 4x^3 + 6x^2 - 4x = (x-1)^4 - 1$, so the equation is $(x-1)^4 - 1 - 2024 = 0$, i.e. $(x-1)^4 = 2025$. Let $y = x - 1$; the four values of y are the fourth roots of 2025, namely $\pm 2025^{1/4}$ and $\pm i 2025^{1/4}$. By symmetry $\sum y_j = \sum y_j^2 = \sum y_j^3 = 0$, while $\sum y_j^4 = 4 \cdot 2025 = 8100$. Then

$$\sum r_j^4 = \sum (1 + y_j)^4 = \sum (1 + 4y_j + 6y_j^2 + 4y_j^3 + y_j^4) = 4 + 0 + 0 + 0 + 8100 = 8104.$$

2. Find the largest positive integer n less than 2026 such that $n^3 - n$ is divisible by 504.

Answer: 2024

Solution: Factor $504 = 2^3 \cdot 3^2 \cdot 7$ and $n^3 - n = (n-1)n(n+1)$. The divisibility conditions are:

- $7 \mid (n-1)n(n+1) \iff n \equiv 0, \pm 1 \pmod{7}$;
- $9 \mid (n-1)n(n+1) \iff n \equiv 0, \pm 1 \pmod{9}$;
- $8 \mid (n-1)n(n+1) \iff n$ is odd, or $8 \mid n$.

Testing $n = 2025$ fails ($2025 \equiv 2 \pmod{7}$). Testing $n = 2024$: $2024 \equiv 1 \pmod{7}$, $2024 \equiv -1 \pmod{9}$, and $8 \mid 2024$. All three conditions hold, so the largest valid n below 2026 is 2024.

3. In triangle ABC , $AB = 14$, $BC = 13$, and $CA = 15$. Let O be the circumcenter and I the incenter of triangle ABC . The value of OI^2 can be expressed as $\frac{m}{n}$ where m and n are relatively prime positive integers. Find $m + n$.

Answer: 129

Solution: With side lengths 13, 14, 15, the semiperimeter is $s = 21$ and the area is $K = \sqrt{21 \cdot 8 \cdot 7 \cdot 6} = 84$. Hence the inradius is $r = \frac{K}{s} = 4$ and the circumradius is $R = \frac{abc}{4K} = \frac{13 \cdot 14 \cdot 15}{336} = \frac{65}{8}$. By Euler's formula relating the two centers,

$$OI^2 = R^2 - 2Rr = \frac{4225}{64} - 2 \cdot \frac{65}{8} \cdot 4 = \frac{4225}{64} - 65 = \frac{65}{64}.$$

Since $\gcd(65, 64) = 1$, we get $m + n = 65 + 64 = 129$.

4. A 10×10 board is to be tiled completely by 25 non-overlapping 1×4 tiles, where each tile may be placed either horizontally or vertically. Prove that no such tiling exists.

Solution: Color the cell in row i , column j with the color $j \bmod 4$. Across columns $1, 2, \dots, 10$ the colors run $1, 2, 3, 0, 1, 2, 3, 0, 1, 2$, so colors 1 and 2 each appear in three columns (30 cells each), while colors 0 and 3 each appear in two columns (20 cells each).

A horizontal 1×4 tile covers four consecutive columns, hence exactly one cell of each color. A vertical 1×4 tile lies in a single column, hence four cells of one color. Let h be the number of horizontal tiles and let v_c be the number of vertical tiles lying entirely in color c . Counting cells of colors 0 and 1,

$$h + 4v_0 = 20, \quad h + 4v_1 = 30.$$

Subtracting gives $4(v_1 - v_0) = 10$, which is impossible since 10 is not divisible by 4. Therefore no tiling exists.